

Research paper

High-order moment interval stability/stabilization and observability of stochastic impulsive T–S fuzzy systems

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ABSTRACT

This article is concerned with the problems of high-order moment interval stability/stabilization and observability of stochastic impulsive T–S fuzzy systems (SITSFSs). Firstly, the definition of high-order moment interval stability is proposed of SITSFSs in this paper. In consideration of the connection between pole placement and performance of systems, sufficient conditions for the distribution of the eigenvalues of systems within a defined interval are provided, serving as the criterion for the interval stability in high-order moment of SITSFSs. Different from the general concept of stability, besides determining the stability in high-order moment of the system, high-order moment interval stability can also describe the dynamic performance in high-order moment of systems. Then, the proposed approach in combination with fuzzy control theory provides a reliable and effective method for designing an interval state feedback controller that can ensure the stability of systems and regulate convergence rate of systems. Furthermore, moment observability of SITSFSs has also been addressed in this article. Finally, the numerical simulation verifies the validity and suitability of the proposed method.

1. Introduction

During normal operation, systems are inevitably disturbed by external factors, such as stochastic [1] and impulsive factors [2]. Stochastic impulsive systems provide a comprehensive framework for characterizing the hybrid evolution of continuous-time and discrete-time dynamics [3]–[4]. Stochastic impulsive systems as a more complex class of systems have a broad range of applications in the area of automation engineering and intelligent control [5]–[6]. Over the last few decades, stability of stochastic systems has been known to be a fundamental concept. And in many cases, it has been investigated. To give a few examples, Luo et al. [7] studied the mean square stability of stochastic systems. The probability dependent stability was investigated in depth in [8]. The articles [9]–[10] explored moment stability, almost deterministic stability, respectively.

Furthermore, it is well known that, in addition to stability, observability is another fundamental issue of a system. In many practical problems, the state variables of a control system are not obtained from direct measurements, but by some observation method. The ability of the results obtained by some observation system to truly reflect the state of the system is the observability of the system. In other words, the major purpose of observability is to evaluate the initial state from outside measurements. In recent years, several articles have delved into the observability of systems. Such as, the observability and controllability were analysed of the systems in [11]. Many of these papers [9–12] in this field focused on the stability and observability of linear systems. In fact,

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a few complex physical models, such as automotive systems and robotic systems, are essentially nonlinear systems. So it is of great interest to investigate the stability and observability of nonlinear stochastic impulsive systems.

Nevertheless, nonlinear characteristics of nonlinear systems are challenging to handle, and many scholars have proposed numerous approaches. For instance, the backstepping method has been used to analyse nonlinear systems in [13]. The authors in [14] have introduced intelligent approximation techniques like fuzzy logic systems (FLS), and Han et al. [15] have employed neural networks (NN) for nonlinear processing. However, none of these approaches are applicable when studying linear systems. Fortunately, Takagi and Sugeno [16] proposed a T-S fuzzy model based on fuzzy logic to dispose of nonlinear systems. The T-S fuzzy model provides a powerful modelling framework for describing nonlinear systems. Its main feature is that the local dynamics of the nonlinear system as several linear models, and through a fuzzy affiliation function the overall fuzzy model is obtained by mixing the local linear model. Up to now, certain foundational questions concerning T-S fuzzy systems (TSFSs) have been well-researched. For example, stability analysis and fuzzy control design problems of TSFSs were investigated in [17–19], and Xie et al. [20] researched stability and how to design affiliation function of discrete-time TSFSs. Mean square stability of TSFSs was investigated in [21], and some relevant studies have been done in [22–24].

However, as far as the authors are aware that most of current articles only think over the mean square stability of TSFSs, i.e., 2nd moment stability, such as [18–21]. The above articles do not consider the higher-order moment stability of TSFSs. Yet high-order moment stability has been broadly applied to practical applications, such as in the financial industry and digital image processing techniques (see [25]). As a result, the examination of higher-order moment stability in TSFSs is a meaningful research focus and serves as the primary motivation for this paper.

It is worth pointing out that many papers have studied the problem by focusing only on the stability of TSFSs, i.e., steady-state performance, and there is little research to conducted on the dynamic properties of TSFSs. Recently, certain academics have introduced new stability criteria based on the idea of pole configuration, which can guarantee the stability of the system while also controlling certain dynamic characteristics of the system, such as convergence speed and response time of the system, and some relevant studies have been done in [26]–[27]. The authors are inspired the above work. Thus, studying the dynamic performance of SITSFSs is the second motivation for this paper. Meanwhile, the problem of high-order moment observability of SITSFSs also has been addressed. This paper succeeds in achieving accurate control over the convergence rate of the systems by cleverly defining high-order moment interval stability. This paper has the following innovations.

(1) This paper is the first to investigate high-order moment interval stability/stabilization and observability of SITSFSs. In contrast to the existing 2^{pth} moment stability criteria, the novel 2^{pth} moment interval stability/stabilization criteria put forward in this paper not only assess the high-order moment stability of the system but also provide an accurate depiction of the dynamic performance of SITSFSs.

(2) Depending on the interval in which the eigenvalues of systems are located, a novel approach to designing a fuzzy controller is given to solve the interval stabilization problem of SITSFSs. The proposed method exhibits that it guarantees both the stability in high-order moment of the system while also enabling accurate regulation the convergence rate of the target system.

Notation: $E(\cdot)$: mathematical expectation arithmetic operator. $\Re(\lambda)$: real part of the eigenvalue λ . $\mathcal{R}^n$ represents the n -dimensional Euclidean space. N : the union of natural numbers. Z_+ : the set of all positive integers. For two integers r_1, r_2 with $r_1 \leq r_2$, the notation $\overline{r_1, r_2}$ represents the set of $r_1, r_1 + 1, \dots, r_2$. For two positive scalars $\tau_0 \geq \tau_1$, $S(\tau_0, \tau_1)$ represents the admissible set of time sequences $\{t_n\}_{n \in N}$ satisfying $0 = t_0 < t_1 < \dots < t_n < \dots, t_n \rightarrow \infty$ for $n \rightarrow \infty$, where $\{t_n\}_{n \in N}$ belong to $S(\tau_0, \tau_1)$, $\tau_0 \leq t_{n+1} - t_n \leq \tau_1$. $\sigma(B)$: the spectrum set of the matrix B . C_{-a}^{-b} : the vertical strip region from $-a$ to $-b$ in the complex plane.

2. Problem description and preliminary preparation

Give a nonlinear stochastic impulsive system and it is able to express as follows:

Rule i: IF $e_1(t)$ is D_1^i ; $e_2(t)$ is D_2^i ;...; $e_g(t)$ is D_g^i , THEN

$$\begin{cases} dx(t) = [A_i x(t) + B_i u(t)]dt + \sum_{l=1}^m G_{li} x(t) dw_l(t), t \neq t_k, \\ x(t_k) = J_i x(t_k^-), t = t_k, \end{cases} \quad (1)$$

where $i \in \overline{1, j}$ and j is the number of rules, $D_e^i (i \in \overline{1, j}, e \in \overline{1, g})$ is the fuzzy set. $e(t)$ stands for the premise variable. $x(t) \in \mathcal{R}^n$ is the state variable. $u(t) \in \mathcal{R}^m$ is the continuously controlled input. w_l is a standard, independent one-dimensional Brownian process, and it is defined on the filtered probability space $(\Omega, \mathcal{F}, P; \mathcal{F}_t)$. Some real constant matrices with appropriate dimensionality A_i, B_i, G_{li}, J_i . The notation $x(t_k) \triangleq x(t_k^+) = \lim_{s \rightarrow 0^+} x(t_k + s)$, and $x(t_k^-) = \lim_{s \rightarrow 0^-} x(t_k + s)$.

Using the common central mean solution fuzzy method, the final SITSFS is obtained as follows

$$\begin{cases} dx(t) = \sum_{i=1}^j \eta_i(e(t)) [(A_i x(t) + B_i u(t))dt + \sum_{l=1}^m G_{li} x(t) dw_l(t)], t \neq t_k, \\ x(t_k) = \sum_{i=1}^j \eta_i(e(t_k)) [J_i x(t_k^-)], t = t_k, \end{cases} \quad (2)$$

where $\eta_i(e(t))$ stands for the normalized membership function, and it is represented as shown below

$$\eta_i(e(t)) = \frac{\beta_i(e(t))}{\sum_{e=1}^j \beta_e(e(t))}, \quad \beta_i(e(t)) = \prod_{e=1}^g D_e^i(e_e(t)),$$

and $D_e^i(\epsilon_e(t)) \geq 0$ stands for the grade of membership of $\epsilon_e(t)$ in D_e^i . Furthermore, it is easy to confirm that

$$\eta_i(\epsilon(t)) \geq 0, \quad \sum_{i=1}^j \eta_i(\epsilon(t)) = 1.$$

Definition 1 ([28]). The zero solution of system (2) is $2p$ th moment asymptotically stable (PMAS) if for any $x_0 \in \mathbb{R}^n$ such that

$$\lim_{t \rightarrow \infty} E \|x(t; t_0, x_0)\|^{2p} = 0, \quad (3)$$

where x_0 is the initial value of system (2).

Definition 2 ([29]). When $k > 0$, vector $v^{(k)}$ is defined as follows

$$\begin{pmatrix} v_1 \\ v_2 \\ \dots \\ v_n \end{pmatrix}^{[k]} = \begin{cases} \begin{pmatrix} v_1(v_1, v_2, \dots, v_n)^{T\{k-1\}} \\ v_2(v_2, v_3, \dots, v_n)^{T\{k-1\}} \\ \dots \\ v_n(v_n)^{T\{k-1\}} \end{pmatrix} \end{cases}$$

Lemma 1 ([3]). For any $v \in \mathbb{R}_n$, there exist an $\frac{(n+p-1)!}{(n-1)!p!} \times n_p$ state transition matrix $\mathcal{H}_{n,p}$ of full column rank, and it is not dependent on v , such that

$$v^{[p]} = \mathcal{H}_{n,p} v^{[p]}.$$

Remark 1. $v^{[p]}$ represents Kronecker product, in which $v^{[p]} = v \otimes v^{[p-1]}$, $p \geq 1$, and $v^{[0]} = 1$. $v^{[p]}$ is a significant extension vector of v for transforming a stochastic system into a deterministic system.

Lemma 2 ([30]). Concern a stochastic system

$$dx(t) = h(t, x(t))dt + j(t, x(t))dw(t),$$

where h and $j \in C^{2,1}(\mathbb{R}^n \times \mathbb{R}_+; \mathbb{R}^n)$. Let $V(t, x) \in C^{2,1}(\mathbb{R}^n \times \mathbb{R}_+; \mathbb{R}^p)$. The stochastic differential of $V(t, x(t))$ is designed by

$$\begin{aligned} dV_l(t, x(t)) = & \left[\frac{\partial V_l}{\partial t}(t, x(t)) + \sum_{m=1}^n \frac{\partial V_l}{\partial x_m} h_m(t, x(t)) + \frac{1}{2} \sum_{m,k=1}^n \frac{\partial^2 V_l}{\partial x_m \partial x_k} j_m(t, x(t)) j_k(t, x(t)) \right] dt \\ & + \sum_{m=1}^n \frac{\partial V_l}{\partial x_m} j_m(t, x(t)) dw(t), l = 1, 2, \dots, p. \end{aligned}$$

Definition 3. System (2) is known as $2p$ th moment asymptotically interval $(-a, -b)$ stable (PMAIS), if system (2) is PMAS and $\sigma(A_{ip}) \subset C_{-a}^{-b} := \{\lambda : -a < \Re(\lambda) < -b\}$, where λ is the eigenvalue with respect to the matrix A_{ip}

$$\begin{aligned} A_{ip} = & (\mathcal{H}_{n,p}^T \mathcal{H}_{n,p})^{-1} \mathcal{H}_{n,p}^T \tilde{A}_{ip} \mathcal{H}_{n,p}, \\ \tilde{A}_{ip} = & \sum_{q=0}^{p-1} I_{n^{p-1-q}} \otimes A_i \otimes I_{n^i} + \sum_{q=0}^{p-2} \sum_{k=0}^q I_{n^{p-2-k}} \otimes \sum_{l=1}^m G_{li} \otimes I_{n^{q-k}} \otimes \sum_{l=1}^m G_{li} \otimes I_{n^k}. \end{aligned}$$

Remark 2. Interval stability is an even more precise stability condition. It not only requires that systems are stable, but also demands that the eigenvalues of states matrices are distributed within a specific region.

Remark 3. When it comes to the control of the system, if parameters a, b are large, the specification of the controller will be large and that will lead to a system that is less resilient to perturbations. By interval stabilization it is possible to keep the controller specification as small as possible, subjecting it to the rate of convergence.

Lemma 3 ([9]). Given $p \in \mathbb{N}$, let $x \in \mathbb{R}^n$ satisfy $\|Ex\|^{2p} < +\infty$, then the inequalities below hold

$$C_0 \|Ex\|^{2p} \leq \|Ex^{(2p)}\| \leq C_1 \|Ex\|^{2p},$$

and

$$\|Ex\|^{2p} \leq E \|x^{(p)}\|^2 \leq C_2 \|Ex\|^{2p},$$

where $C_0 = \frac{1}{\sqrt{n^{2p-1}}}$, $C_1 = d_{n,2p}$, $C_2 = d_{n,p}$.

Remark 4. In this paper, the original SITSFS is able to obtained as an extended system model described by SDE through a series of techniques like the $\mathcal{H}$ -representation approach. It is sufficient to show that when the extended system described by SDE is mean square asymptotic interval stability, the original system is $2p$ th moment asymptotic interval stability.

3. 2pth moment interval stability/stabilization

Within this part, our focus will be on examining 2pth moment asymptotic interval stability/stabilization. Letting $u(t) = 0$ of system (2), the following system is available

$$\begin{cases} dx(t) = \sum_{i=1}^j \eta_i(\epsilon(t))[(A_i x(t))dt + \sum_{l=1}^m G_{li} x(t)dw_l(t)], t \neq t_k, \\ x(t_k) = \sum_{i=1}^j \eta_i(\epsilon(t_k))[J_i x(t_k^-)], t = t_k. \end{cases} \quad (4)$$

When $k = 2$, apply the $I\hat{o}$ formula to the vector $x^{[k]}$ using Definition 2 and Remark 2. It follows as

$$\begin{aligned} & dx^{[2]}(t) \\ &= dx(t) \otimes x(t) + x(t) \otimes dx(t) + dx(t) \otimes dx(t) \\ &= \sum_{i=1}^j \eta_i(\epsilon(t))[A_i x(t)dt + \sum_{l=1}^m G_{li} x(t)dw_l(t)] \otimes x(t) + x(t) \otimes \sum_{i=1}^j \eta_i(\epsilon(t))[A_i x(t)dt + \sum_{l=1}^m G_{li} x(t)dw_l(t)] \\ &\quad + \sum_{i=1}^j \eta_i(\epsilon(t))[A_i x(t) + \sum_{l=1}^m G_{li} x(t)dw_l(t)] \otimes \sum_{i=1}^j \eta_i(\epsilon(t))[A_i x(t)dt + \sum_{l=1}^m G_{li} x(t)dw_l(t)] \\ &= (\sum_{i=1}^j \eta_i(\epsilon(t))A_i \otimes I)x^{[2]}(t)dt + (I \otimes \sum_{i=1}^j \eta_i(\epsilon(t))A_i)x^{[2]}(t)dt + (\sum_{i=1}^j \eta_i(\epsilon(t)) \sum_{l=1}^m G_{li} x(t) \\ &\quad \otimes \sum_{i=1}^j \eta_i(\epsilon(t)) \sum_{l=1}^m G_{li} x(t))x^{[2]}(t)dt + (\sum_{i=1}^j \eta_i(\epsilon(t)) \sum_{l=1}^m G_{li} \otimes I + I \otimes \sum_{i=1}^j \eta_i(\epsilon(t)) \sum_{l=1}^m G_{li})x^{[2]}(t)dw_l(t). \end{aligned}$$

Integrating either side of the equation

$$\begin{aligned} & x(t) \otimes x(t) \\ &= x_0 \otimes x_0 + \int_0^t [(\sum_{i=1}^j \eta_i(\epsilon(t))A_i \otimes I)x^{[2]}(s) + (I \otimes \sum_{i=1}^j \eta_i(\epsilon(t))A_i)x^{[2]}(s) + (\sum_{i=1}^j \eta_i(\epsilon(t)) \sum_{l=1}^m G_{li} \\ &\quad \otimes \sum_{i=1}^j \eta_i(\epsilon(t)) \sum_{l=1}^m G_{li})x^{[2]}(s)]ds + \int_0^t (\sum_{i=1}^j \eta_i(\epsilon(t)) \sum_{l=1}^m G_{li} \otimes I + I \otimes \sum_{i=1}^j \eta_i(\epsilon(t)) \sum_{l=1}^m G_{li})x^{[2]}(s)dw_l(s). \end{aligned}$$

Set $X^{[2]} = E[x^{[2]}]$. Using Lemma 1 as a basis, we can determine the expected value of the aforementioned equation

$$\begin{aligned} & \mathcal{H}_{n,1} X^{[2]}(t) \\ &= \mathcal{H}_{n,1} X_0^{[2]} + \int_0^t [(\sum_{i=1}^j \eta_i(\epsilon(t))A_i \otimes I) + (I \otimes \sum_{i=1}^j \eta_i(\epsilon(t))A_i) + (\sum_{i=1}^j \eta_i(\epsilon(t)) \sum_{l=1}^m G_{li} \otimes I \\ &\quad + I \otimes \sum_{i=1}^j \eta_i(\epsilon(t)) \sum_{l=1}^m G_{li})] \times \mathcal{H}_{n,1} X^{[2]}(s)ds. \end{aligned}$$

Deriving both sides of the aforementioned equation with respect to t results in

$$\begin{aligned} & \mathcal{H}_{n,1} \dot{X}^{[2]}(t) \\ &= [(\sum_{i=1}^j \eta_i(\epsilon(t))A_i \otimes I) + (I \otimes \sum_{i=1}^j \eta_i(\epsilon(t))A_i) + (\sum_{i=1}^j \eta_i(\epsilon(t)) \sum_{l=1}^m G_{li} \otimes I + I \otimes \sum_{i=1}^j \eta_i(\epsilon(t)) \sum_{l=1}^m G_{li})] \times \mathcal{H}_{n,1} X^{[2]}(t). \end{aligned}$$

By multiplying both the left-hand and right-hand sides of the given expression by $(\mathcal{H}_{n,1}^T \mathcal{H}_{n,1})^{-1} \mathcal{H}_{n,1}^T$ simultaneously yields

$$\begin{aligned} \dot{X}^{[2]}(t) &= (\mathcal{H}_{n,1}^T \mathcal{H}_{n,1})^{-1} \mathcal{H}_{n,1}^T [(\sum_{i=1}^j \eta_i(\epsilon(t))A_i \otimes I) + (I \otimes \sum_{i=1}^j \eta_i(\epsilon(t))A_i) + (\sum_{i=1}^j \eta_i(\epsilon(t)) \sum_{l=1}^m G_{li} \\ &\quad \otimes \sum_{i=1}^j \eta_i(\epsilon(t)) \sum_{l=1}^m G_{li})] \times \mathcal{H}_{n,1} X^{[2]}(t). \end{aligned}$$

By mathematical induction it follows that

$$\begin{cases} \dot{X}^{[p]}(t) = \sum_{i=1}^j \eta_i(\epsilon(t))A_{ip} X^{[p]}(t), t \neq t_k, \\ X^{[p]}(t_k) = \sum_{i=1}^j \eta_i(\epsilon(t_k))J_{ip} X^{[p]}(t_k^-), t = t_k, \end{cases} \quad (5)$$

where

$$\begin{aligned} A_{ip} &= (\mathcal{H}_{n,p}^T \mathcal{H}_{n,p})^{-1} \mathcal{H}_{n,p}^T \tilde{A}_{ip} \mathcal{H}_{n,p}, \\ \tilde{A}_{ip} &= \sum_{q=0}^{p-1} I_{n^{p-1-q}} \otimes A_i \otimes I_{n^q} + \sum_{q=0}^{p-2} \sum_{k=0}^q I_{n^{p-2-k}} \otimes \sum_{l=1}^m G_{li} \otimes I_{n^{q-k}} \otimes \sum_{l=1}^m G_{li} \otimes I_{n^k}, \\ J_{ip} &= (\mathcal{H}_{n,p}^T \mathcal{H}_{n,p})^{-1} \mathcal{H}_{n,p}^T J_i^{[p]} \mathcal{H}_{n,p}. \end{aligned}$$

3.1. 2pth moment interval stability

Initially, the high-order moment interval stability of system (4) is considered. In this section, a time-varying discrete Lyapunov function will be established to establish some stability criteria for system (4) with respect to impulse time sequences. For that reason, split the impulsive interval $[t_k, t_{k+1})$, $k \in N$ into Z_+ subintervals $\Delta_{kn} \triangleq [t_k + rh_k, t_k + (r+1)h_k)$, $r \in S_{Z_+} \triangleq \{0, 1, \dots, Z_+ - 1\}$, in which $h_k = \frac{t_{k+1} - t_k}{Z_+}$. Then, some piecewise linear functions are defined as follows

$$\phi_{0r}(t) = \frac{t_k + (r+1)h_k - t}{h_k}, t \in \Delta_{kr}, n \in S_{Z_+}, k \in N.$$

Set $\phi_{1r}(t) = 1 - \phi_{0r}(t)$. It is simple to see, for any $t \in R_+$, $\frac{1}{h_k} = Z_+ \delta(t)$, $\phi_{0r}(t) \in [0, 1]$, $\phi_{1r}(t) \in [0, 1]$, and

$$\phi_{0r}(t_k + rh_k) = \phi_{1r}((t_k + (r+1)h_k)^-) = \xi(t_k^-) = 1,$$

$$\phi_{0r}((t_k + (r+1)h_k)^-) = \phi_{1r}(t_k + rh_k) = \xi(t_k) = 0, \quad (6)$$

where $\delta(t) = \frac{1}{t_{k+1} - t_k}$, $\xi(t) = (t - t_k)\delta(t)$. For any impulse sequence $\{t_k\}_{k \in N} \in S(\tau_0, \tau_1)$, there is $\theta_0(t): R_+ \rightarrow [0, 1]$ such that

$$\delta(t) = \frac{\theta_0(t)}{\tau_0} + \frac{\theta_1(t)}{\tau_1},$$

where $\theta_1(t) = 1 - \theta_0(t)$ and $\theta_0(t) = \frac{\delta(t) - \frac{1}{\tau_1}}{\frac{1}{\tau_0} - \frac{1}{\tau_1}}$, if $\tau_0 \neq \tau_1$; else, $\theta_0(t) = 1$.

By thinking about the phenomenon of system (4) jumping, the piecewise Lyapunov function is introduced as

$$V(t) = \mu^{\xi(t)} X^{[p]}(t)^T P(t) X^{[p]}(t), \quad (7)$$

where $\mu > 0$, $P(t) = \sum_{r=0}^{Z_+-1} P_r(t)$, $P_r(t) = \sum_{s=0}^1 \phi_{sr}(t) P_{r+s}$, $t \in \Delta_{kn}$, $r \in S_{Z_+}$ with $P_r > 0$, $\bar{r} = 0, 1, 2, \dots, Z_+$. Demonstrating the continuity of $V(t)$ within each impulse time interval (t_k, t_{k+1}) , $k \in N$ is a simple task.

Remark 5. On the foundation of the extended system model, the impulsive interval is partitioned so as to construct a time-varying discrete Lyapunov function, which theoretical analysis and numerical calculations show to be an effective method for reducing conservatism.

Theorem 1. Given a class $S(\tau_0, \tau_1)$ of impulse sequences, positive integer Z_+ , positive real numbers a, b ($a > b$), γ, p, μ , if there exist $n \times n$ matrices $P_r > 0$, $\bar{r} = 0, 1, \dots, Z_+$ satisfying

$$\Psi_{rsc} \triangleq \frac{\ln(\mu)}{\tau_c} P_{r+s} + \frac{N}{\tau_c} (P_{r+1} - P_r) + P_{r+s} [-(A_{ip} + aI)] - [(A_{ip} + aI)]^T P_{r+s} + \gamma P_{r+s} < 0, \quad (8)$$

$$\Phi_{rsc} \triangleq \frac{\ln(\mu)}{\tau_c} P_{r+s} + \frac{N}{\tau_c} (P_{r+1} - P_r) + P_{r+s} (A_{ip} + bI) + (A_{ip} + bI)^T P_{r+s} + \gamma P_{r+s} < 0, \quad (9)$$

$$J_{ip}^T P_0 J_{ip} - P_N < 0, \quad (10)$$

where $r \in S_{Z_+}$, $s, c = 0, 1$, A_{ip} , J_{ip} are defined in system (5), then system (4) is 2pth moment asymptotically interval $(-a, -b)$ stable over $S(\tau_0, \tau_1)$.

Proof. It follows from Lemma 2 and Remark 4 that if one proves that system (4) is 2pth moment interval stable, then one need only prove that system (5) is mean square interval stable. By Definitions 1 and 4, system (5) is mean square asymptotically interval stable and $\sigma(A_{ik}) \in (-a, -b)$, if (8) – (10) hold. Therefore, if both following systems

$$\begin{cases} \dot{X}^{[p]}(t) = - \sum_{i=1}^j \eta_i(e(t)) (A_{ip} + aI) X^{[p]}(t), t \neq t_k, \\ X^{[p]}(t_k) = \sum_{i=1}^j \eta_i(e(t)) J_{ip} X^{[p]}(t_k^-), t = t_k, \end{cases} \quad (11)$$

and

$$\begin{cases} \dot{X}^{(p)}(t) = \sum_{i=1}^j \eta_i(\epsilon(t))(A_{ip} + bI)X^{(p)}(t), t \neq t_k, \\ X^{(p)}(t_k) = \sum_{i=1}^j \eta_i(\epsilon(t))J_{ip}X^{(p)}(t_k^-), t = t_k. \end{cases} \quad (12)$$

For simplicity of notation presentation, set $\Gamma_r(t) \triangleq \sum_{s,c \in \mathbb{N}}^1 \phi_{sr}(t)\theta_c(t)\Psi_{rsc}$. For (8), it follows that

$$\Gamma_r(t) < 0, r \in S_{Z_+}, \quad (13)$$

for (13), if $x_0 = 0$, then $x(t) \triangleq x(t, t_0, x_0) \equiv 0$. Thus, for $x_0 \neq 0$, the stopping time $\varsigma = \inf\{t > t_0 \mid x_t = 0\}$ is defined, that describes exactly the solution corresponding to zero for the first time. It is known that $\varsigma = +\infty$ if J_{ik} are invertible matrices. A power time-dependent discretized Lyapunov function is introduced, i.e., $\widehat{W}(t) = e^{pr(t-t_0)}[V(t)]^p$, where $V(t)$ defined in (7). Deriving $V(t)$ with respect to t , for $t \in [t_k + rh_k, t_k + (r+1)h_k]$ with any given $k \in N_0$, and $r \in S_{Z_+}$ can result in

$$\begin{aligned} \dot{V}(t) = & (X^{(p)})^T \{ \ln(\mu)\delta(t)P_r(t) + N\delta(t)(P_{r+1} - P_r) + P_r(t)[- \sum_{i=1}^k \eta_i(\epsilon(t))(A_{ip} + aI)] \\ & + [- \sum_{i=1}^k \eta_i(\epsilon(t))(A_{ip} + aI)]^T P_r(t) \} X^{(p)}. \end{aligned}$$

Similarly for $\widehat{W}(t)$ the derivative, for $t \in [t_k + rh_k, t_k + (r+1)h_k \wedge \varsigma)$ can access

$$\dot{\widehat{W}}(t) = p\nu^{p-1}e^{r(t-t_0)}[\dot{V}(t) + \gamma V(t)](t) = p\nu^{p-1}e^{r(t-t_0)}[(X^{(p)})^T(t)\Gamma_r(t - rh_k)(X^{(p)})](t) \leq 0.$$

Integrate both sides of the above inequality from $t_k + rh_k$ to $t \wedge \varsigma$, the inequality below can be derived

$$\widehat{W}(t \wedge \varsigma) \leq \widehat{W}(t_k + rh_k). \quad (14)$$

Since $V(t)$ is continuous on $[t_k, t_{k+1})$, $\widehat{W}(t)$ is also continuous on $[t_k, t_{k+1})$. Then, for any $t \in [t_k, t_{k+1} \wedge \varsigma)$, inequality can be deduced into

$$\widehat{W}(t \wedge \varsigma) \leq \widehat{W}(t_k). \quad (15)$$

Next, $\widehat{W}(t)$ at impulsive instant $t_k, k \in N_0$ will be estimated. From the definition of $P_r(t)$ can gain

$$P_0(t_k) = P_0, \quad P_{N-1}(t_k^-) = P_N. \quad (16)$$

Then, using (10), for any $k \in N_0$, we can get

$$\begin{aligned} V(t_k) &= \mu^{\xi(t)}(X^{(p)})^T(t_k)P_0X^{(p)}(t_k) \\ &= \mu^{\xi(t)} \sum_{i=1}^j \eta_i(\epsilon(t))(X^{(p)})^T(t_k^-)(J^{(ip)})^T P_0 \sum_{i=1}^j \eta_i(\epsilon(t))J^{(ip)}X^{(p)}(t_k^-) \\ &\leq \mu^{\xi(t)}(X^{(p)})^T(t_k^-)P_NX^{(p)}(t_k^-) = V(t_k^-), \end{aligned}$$

whence $\widehat{W}(t_k) \leq \widehat{W}(t_k^-)$. By applying this inequality jointly and (15) can receive $\widehat{W}(t \wedge \varsigma) \leq \widehat{W}(t_0), t \geq t_0$. So we have an estimate

$$\|x^{(p)}(t)\|^{2p} \leq Q\|x^{(p)}(t_0)\|^{2p}e^{-pr(t-t_0)}, t \geq t_0. \quad (17)$$

Thus, $\lim_{t \rightarrow \infty} E\|x(t; t_0, x_0)\|^{2p} = 0$ can be obtained, where Q is a positive scalar. Then system (5) is mean square asymptotically stable over $S(\tau_0, \tau_1)$. It shows that system (4) is PMAS, while we can also get $\sigma(A_i) \in C_{-a}^{+\infty}$. Similarly, it is easy to show stable criterion (9) of system (12), and system (4) is proved to be PMAS and $\sigma(A_i) \in C_{-\infty}^{-b}$. Therefore, system (4) is PMAIS. The proof is complete.

The conditions in [Theorem 1](#) can be weakened to obtain the sufficient condition for system (4) as PMAS, as shown below.

Corollary 1. Given a class $S(\tau_0, \tau_1)$ of impulse sequences, positive integer Z_+ , positive real numbers γ, p, μ , if there exist $n \times n$ matrices $P_r > 0, \bar{r} = 0, 1, \dots, Z_+$ satisfying

$$\begin{aligned} \Psi_{rsc} &\triangleq \frac{\ln(\mu)}{\tau_c} P_{r+s} + \frac{N}{\tau_c} (P_{r+1} - P_r) + P_{r+s} A_{ip} + (A_{ip})^T P_{r+s} + \gamma P_{r+s} < 0, r \in S_{Z_+}, s, c = 0, 1, \\ J_{ip}^T P_0 J_{ip} - P_N &< 0, \end{aligned}$$

where the definition of A is given in [Theorem 1](#), then system (4) is $2p$ th moment asymptotically stable over $S(\tau_0, \tau_1)$.

3.2. 2pth moment interval stabilization

Moving forward, our investigation will centre on the stability of SITSFSs, and designing a fuzzy state feedback controller controls the rate convergence of SITSFSs,

$$u(t) = K_i x(t), i = 1, 2, \dots, j,$$

where K_i is constant gain matrix. Define $K(\eta) = \sum_{i=1}^j \eta_i(\epsilon(t)) K_i$, the overall fuzzy controller can be written as

$$u(t) = \sum_{i=1}^j \eta_i(\epsilon(t)) K_i x(t) = K_\eta x(t). \quad (18)$$

Substituting (18) into system (2), we can obtain

$$\begin{cases} \dot{x}(t) = \sum_{i=1}^j \sum_{l=1}^j \eta_l(\epsilon(t)) [(A_i + B_i K_i) x(t)](t), t \neq t_k, \\ x(t_k) = \sum_{i=1}^j \eta_i(\epsilon(t_k)) [J_{ip} x(t_k^-)], t = t_k. \end{cases} \quad (19)$$

Similarly, the corresponding extended system for system (19) can be obtained as

$$\begin{cases} \dot{X}^{[p]}(t) = \sum_{i=1}^j \eta_i(\epsilon(t)) \underline{A}_{ip} X^{[p]}(t), t \neq t_k, \\ X^{[p]}(t_k) = \sum_{i=1}^j \eta_i(\epsilon(t)) J_{ip} X^{[p]}(t_k^-), t = t_k, \end{cases} \quad (20)$$

where

$$\begin{aligned} \underline{A}_{ip} &= (\mathcal{H}_{n,p}^T \mathcal{H}_{n,p})^{-1} \mathcal{H}_{n,p}^T \check{A}_{ip} \mathcal{H}_{n,p}, \\ \check{A}_{ip} &= \sum_{q=0}^{p-1} I_{n^{p-1-q}} \otimes (A_i + B_i K_i) \otimes I_{n^q} + \sum_{q=0}^{p-2} \sum_{k=0}^q I_{n^{p-2-k}} \otimes \sum_{l=1}^m G_{li} \otimes I_{n^{q-k}} \otimes \sum_{l=1}^m G_{li} \otimes I_{n^k}, \\ J_{ip} &= (\mathcal{H}_{n,p}^T \mathcal{H}_{n,p})^{-1} \mathcal{H}_{n,p}^T J_i^{[p]} \mathcal{H}_{n,p}. \end{aligned}$$

Theorem 2. Given a class $S(\tau_0, \tau_1)$ of impulse sequences, positive integer Z_+ , positive real numbers a, b ($a > b$), γ, p, μ , if there exist $n \times n$ matrices $P_r > 0, \bar{r} = 0, 1, \dots, Z_+$ satisfying

$$Y_{rsc} \triangleq \frac{\ln(\mu)}{\tau_c} P_{r+s} + \frac{N}{\tau} (P_{r+1} - P_r) + P_{r+s} \underline{A}_{1ip} + \underline{A}_{1ip}^T P_{r+s} p + \gamma P_{r+s} < 0, r \in S_{Z_+}, s, c = 0, 1 \quad (21)$$

$$\Theta_{rsc} \triangleq \frac{\ln(\mu)}{\tau_c} P_{r+s} + \frac{N}{\tau} (P_{r+1} - P_r) + P_{r+s} \underline{A}_{2ip} + \underline{A}_{2ip}^T P_{r+s} p + \gamma P_{r+s} < 0, r \in S_{Z_+}, s, c = 0, 1 \quad (22)$$

$$J_{ip}^T P_0 J_{ip} - P_N < 0, \quad (23)$$

where

$$\begin{aligned} \underline{A}_{1ip} &= (\mathcal{H}_{n,p}^T \mathcal{H}_{n,p})^{-1} \mathcal{H}_{n,p}^T \check{A}_{1ip} \mathcal{H}_{n,p}, \\ \check{A}_{1ip} &= \sum_{q=0}^{p-1} I_{n^{p-1-q}} \otimes [-(A_i + aI)] - B_i K_i \otimes I_{n^q} + \sum_{q=0}^{p-2} \sum_{k=0}^q I_{n^{p-2-k}} \otimes \sum_{l=1}^m G_{li} \otimes I_{n^{q-k}} \otimes \sum_{l=1}^m G_{li} \otimes I_{n^k}, \\ \underline{A}_{2ip} &= (\mathcal{H}_{n,p}^T \mathcal{H}_{n,p})^{-1} \mathcal{H}_{n,p}^T \check{A}_{2ip} \mathcal{H}_{n,p}, \\ \check{A}_{2ip} &= \sum_{q=0}^{p-1} I_{n^{p-1-q}} \otimes [(A_i + bI)] + B_i K_i \otimes I_{n^q} + \sum_{q=0}^{p-2} \sum_{k=0}^q I_{n^{p-2-k}} \otimes \sum_{l=1}^m G_{li} \otimes I_{n^{q-k}} \otimes \sum_{l=1}^m G_{li} \otimes I_{n^k}, \\ J_{ip} &= (\mathcal{H}_{n,p}^T \mathcal{H}_{n,p})^{-1} \mathcal{H}_{n,p}^T J_i^{[p]} \mathcal{H}_{n,p}, \end{aligned}$$

then system (2) is 2pth moment interval stabilizable over $S(\tau_0, \tau_1)$.

Proof. The proof follows a similar process as in Theorem 1, and the details are omitted for brevity.

Now give the exact steps to solve for K .

$$\underline{A}_{1ip} = (\mathcal{H}_{n,p}^T \mathcal{H}_{n,p})^{-1} \mathcal{H}_{n,p}^T \check{A}_{1ip} \mathcal{H}_{n,p},$$

$$\begin{aligned}
\check{A}_{1ip} &= \sum_{q=0}^{p-1} I_{n^{p-1-q}} \otimes [-(A_i + aI)] - B_i K_i \otimes I_{n^q} + \sum_{q=0}^{p-2} \sum_{k=0}^q I_{n^{p-2-k}} \otimes \sum_{l=1}^m G_{li} \otimes I_{n^{q-k}} \otimes \sum_{l=1}^m G_{li} \otimes I_{n^k}, \\
&= \sum_{q=0}^{p-1} I_{n^{p-1-q}} \otimes [-(A_i + aI)] \otimes I_{n^q} + \sum_{q=0}^{p-2} \sum_{k=0}^q I_{n^{p-2-k}} \otimes \sum_{l=1}^m G_{li} \otimes I_{n^{q-k}} \otimes \sum_{l=1}^m G_{li} \otimes I_{n^k} \\
&\quad - \sum_{q=0}^{p-1} I_{n^{p-1-q}} \otimes ([B_i K_i]) \otimes I_{n^q}, \\
&= \sum_{q=0}^{p-1} I_{n^{p-1-q}} \otimes [-(A_i + aI)] \otimes I_{n^q} + \sum_{q=0}^{p-2} \sum_{k=0}^q I_{n^{p-2-k}} \otimes \sum_{l=1}^m G_{li} \otimes I_{n^{q-k}} \otimes \sum_{l=1}^m G_{li} \otimes I_{n^k} \\
&\quad - \sum_{q=0}^{p-1} (I_{n^{p-1-q}} \otimes B_i \otimes I_{n^q})(I_{n^{p-1-q}} \otimes K_i \otimes I_{n^q}).
\end{aligned}$$

(i) Setting $W_i = H_{n,p}(I_{n^{p-1-i}} \otimes K_i \otimes I_{n^i})P_{r+s}^{-1}$, the above linear equation for matrix W_i is obtained.

(ii) By finding the basic solutions of the above system of linear equations, then we can obtain a linear representation of W_i .

(iii) Substituting a linear representation of W_i into (21) or (22), K_i is available.

The conditions in [Theorem 2](#) can be weakened to obtain the sufficient condition for system (2) as $2p$ th moment stabilizable, as shown below.

Corollary 2. Given a class $S(\tau_0, \tau_1)$ of impulse sequences, positive integer Z_+ , γ, p, μ , if there exist $n \times n$ matrices $P_r > 0$, $\bar{r} = 0, 1, \dots, Z_+$ satisfying

$$Y_{rsc} \triangleq \frac{\ln(\mu)}{\tau_c} P_{r+s} + \frac{N}{\tau_c} (P_{r+1} - P_r) + P_{r+s} \underline{A}_{ip} + \underline{A}_{ip}^T P_{r+s} + \gamma P_{r+s} < 0, r \in S_{Z_+}, s, c = 0, 1,$$

$$J_{ip}^T P_0 J_{ip} - P_N < 0,$$

where

$$\underline{A}_{ip} = (H_{n,p}^T H_{n,p})^{-1} H_{n,p}^T \check{A}_{ip} H_{n,p},$$

$$\check{A}_{1ip} = \sum_{q=0}^{p-1} I_{n^{p-1-q}} \otimes (A_i - B_i K_i) \otimes I_{n^q} + \sum_{q=0}^{p-2} \sum_{k=0}^q I_{n^{p-2-k}} \otimes \sum_{l=1}^m G_{li} \otimes I_{n^{q-k}} \otimes \sum_{l=1}^m G_{li} \otimes I_{n^k},$$

$$J_{ip} = (H_{n,p}^T H_{n,p})^{-1} H_{n,p}^T J_i^{[p]} H_{n,p},$$

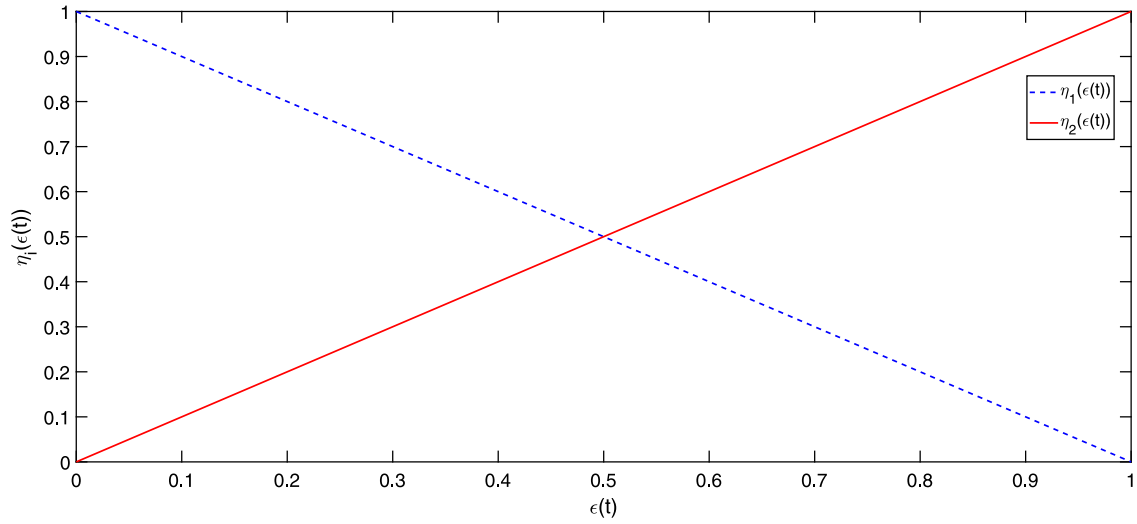
then system (2) is $2p$ th moment stabilizable over $S(\tau_0, \tau_1)$.

Remark 6. As per the author's understanding, current approaches to high-order moment stability of SITSFSs do not address the convergence speed as the system nears equilibrium. The novel method proposed in this paper for high-order moment interval stabilization overcomes this limitation. In contrast to existing high-order moment stabilization techniques, [Theorem 2](#) offers a more precise control over the convergence rate of the system in high-order moment. By adjusting the parameters a and b , the convergence speed of the system's states in high-order moment can be tailored to the desired level.

4. $2p$ th moment observability

The observability of a system is another fundamental concept in systems theory. It is a measure of the ability to estimate the current state of a system from external outputs (such as metrics and tracking). In other words, a system is observable if you can look outside of it to determine what is happening inside of it. It has been studied in a range of cases; see [\[31–33\]](#). Consider the SITSFS with output and zero input

$$\begin{cases} dx(t) = \sum_{i=1}^j \eta_i(e(t)) [A_i x(t) dt + \sum_{l=1}^m G_{li} x(t) dw_l(t)], t \neq t_k, \\ x(t_k) = \sum_{i=1}^j \eta_i(e(t_k)) [J_i x(t_k^-)], t = t_k, \\ y(t) = \sum_{i=1}^j \eta_i(e(t)) R_i x(t), \end{cases} \quad (24)$$

Fig. 1. Membership functions $\eta_i(\epsilon(t))$.

where A_i , $\sum_{l=1}^m G_{li}$ are defined as in system (1) and R_i denotes a matrix of suitable dimensions. In the same way as the above work, system (24) can be described as the following extended system

$$\begin{cases} \dot{X}^{[p]}(t) = \sum_{i=1}^j \eta_i(\epsilon(t)) A_{ip} X^{[p]}(t), t \neq t_k, \\ X^{[p]}(t_k) = \sum_{i=1}^j \eta_i(\epsilon(t)) J_{ip} X^{[p]}(t_k^-), t = t_k, \\ y^p(t) = \sum_{i=1}^j \eta_i(\epsilon(t)) R_{ip} X^{[p]}(t), \end{cases} \quad (25)$$

where A_{ip} , J_{ip} are defined above. $R_{ip} = (H_{n,p}^T H_{n,p})^{-1} H_{n,p}^T R_i^{[p]} H_{n,p}$.

Definition 4. System (24) is known to be $2p$ th moment observable at t_0 , if there is a finite interval of time $[t_0, t_s]$, such that when $E[y^{2p}] \equiv 0$, $\forall t \in [t_0, t_s]$, one has $E[x^{2p}(t_0)] = 0$.

Remark 7. High-order moment observability is similar in nature to high-order moment stability, i.e. higher-order moment observability guarantees lower-order moment observability.

Theorem 3. Let $i(t, t_0) = A_{ip} i(t, t_0)$, $i(t, t_0) = I \frac{(k+n-1)!}{(n-1)!k!} \times \frac{(k+n-1)!}{(n-1)!k!}$, then system (24) is $2p$ th moment observable at time t_0 if there is some interval $[t_0, t_1]$ with $t_1 > t_0$ satisfying one of the following

- (1) The Grammian matrix $\Xi[t_0, t_1] = \int_{t_0}^{t_1} i^T(t, t_0) R_{ip}^T R_{ip} i(t, t_0) dt$ is nonsingular.
- (2) $\Xi[t_0, t_1]$ is positive definite.

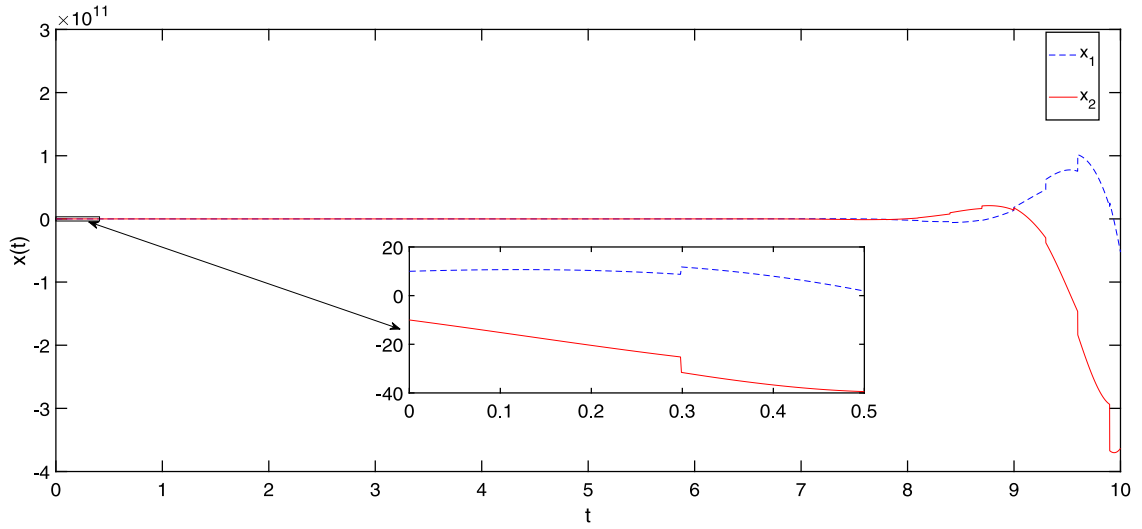
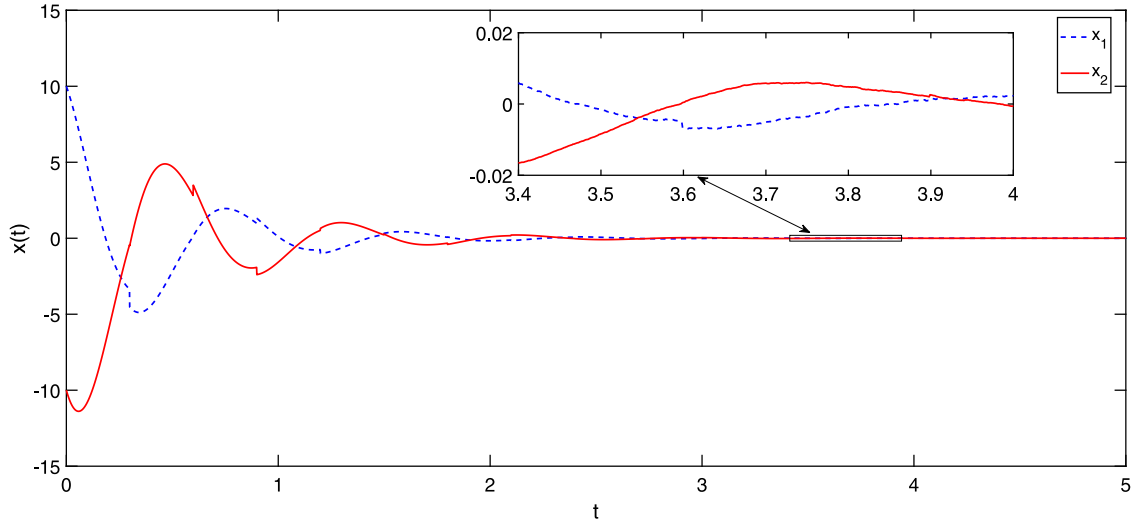
Proof. System (24) is $2p$ th moment observable at time t_0 if system (25) is observable at time t_0 , i.e., condition (1) or condition (2) in Definition 3 holding.

Since $i(t, t_0)$ in Theorem 3 is not easily computed, we give an easy to calculate criteria as follows.

Corollary 3. The system (24) is $2p$ th moment completely observable if $\text{rank}(\Delta) = \frac{(k+n-1)!}{(n-1)!k!}$, where $\Delta \triangleq [R_{ip}^T (R_{ip} A_{ip})^T \cdots (R_{ip} A_{ip}^{\frac{(k+n-1)!}{(n-1)!k!} - 1})^T]^T$ is known as observable matrix.

5. Numerical simulation

In this section, the proposed method has been demonstrated to be applicable and effective by means of a numerical simulation example (see Ref. [17], and other related literature) of precise control.

Fig. 2. State responses under $u(t) = 0(p = 1)$.Fig. 3. States responses ($p = 1$) with controller.

Ponder over the nonlinear stochastic system with impulses as described

$$dx_1(t) = (3 + 0.1\cos^2(x_1(t)))x_1(t) + 2x_2(t) - u_1(t) - 0.5u_2(t) + [0.03x_1(t) + 0.02x_2(t)]dw(t),$$

$$dx_2(t) = (-5 + 0.2\cos^2(x_1(t)))x_1(t) + 0.1\cos^2(x_2(t)) + 2u_1(t) - u_2(t) + [(-0.024x_1(t)) + (-0.01x_2(t))]dw(t),$$

$$dx_1(t_k) = (1.3 + 0.01\cos^2(x_1(t_k^-)))x_1(t_k^-),$$

$$dx_2(t_k) = 1.25x_2(t_k^-).$$

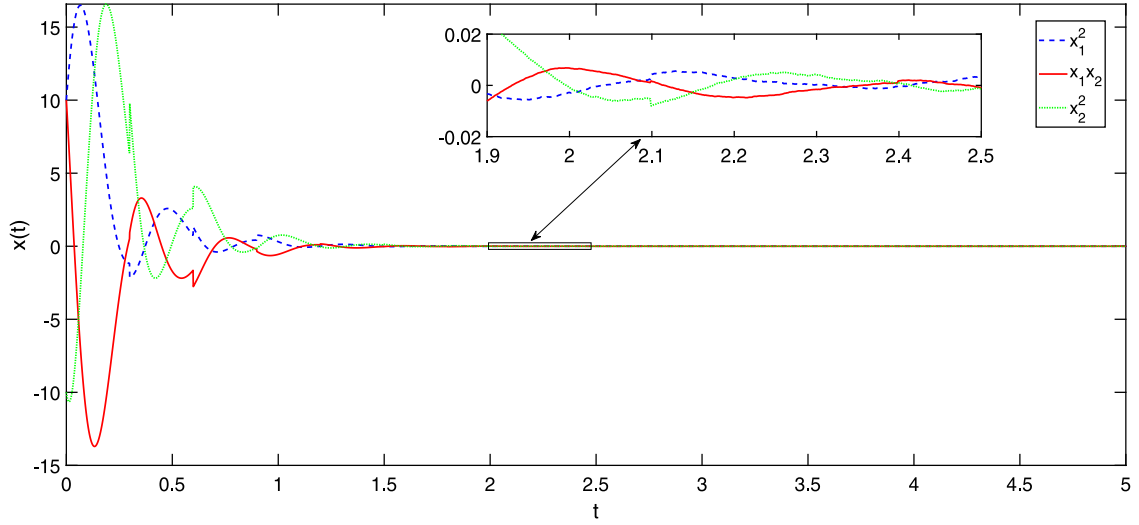
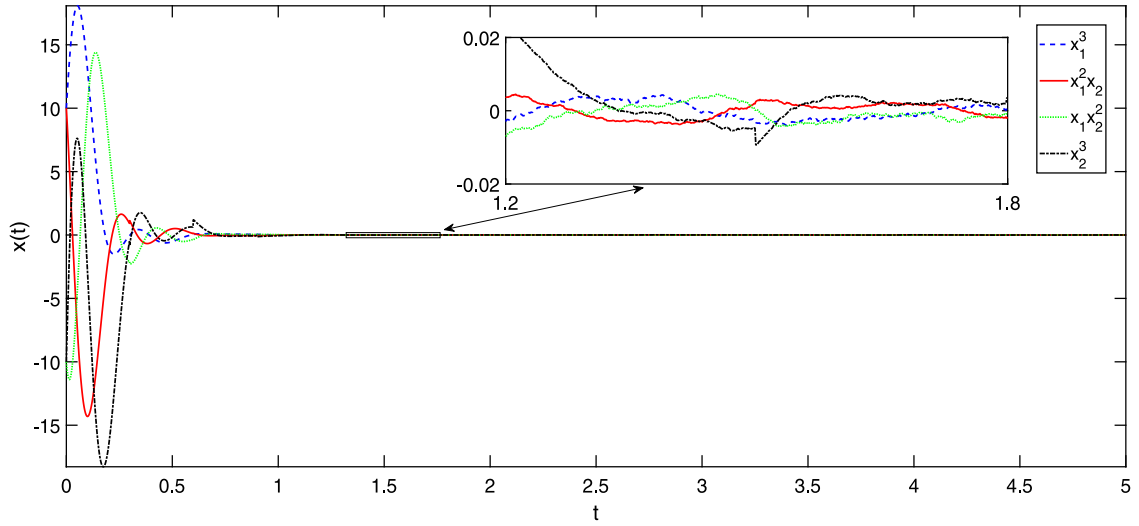
(26)

Let $\epsilon(t) = \cos^2(x_1(t))$ and $x(t) = [x_1(t), x_2(t)]^T$. It is possible to depict system (26) using the T-S fuzzy model that incorporates two rules

Rule 1: IF $\epsilon(t)$ is $\frac{\pi}{2}$, THEN

$$dx(t) = A_1x(t) + Bu(t) + G_{w_1}dw(t),$$

$$x(t_k) = J_1x(t_k^-), t = t_k, k \in \mathbb{Z}^+.$$

Fig. 4. States responses ($p = 2$) with controller.Fig. 5. States responses ($p = 3$) with controller.

Rule 2: IF $\epsilon(t)$ is 0, THEN

$$dx(t) = A_2x(t) + Bu(t) + G_{w_2}dw(t),$$

$$x(t_k) = J_2x(t_k^-), t = t_k, k \in \mathbb{Z}^+.$$

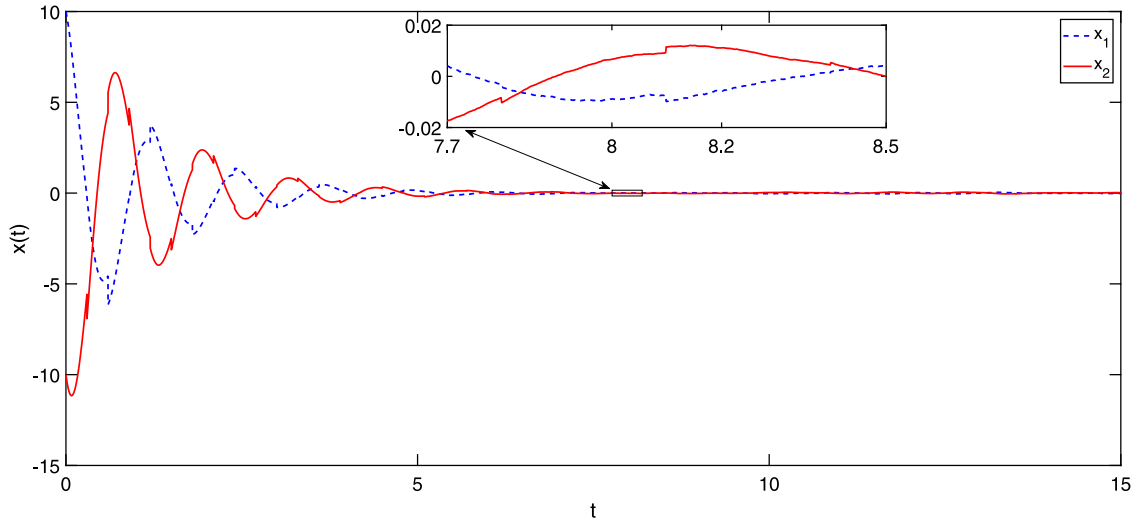
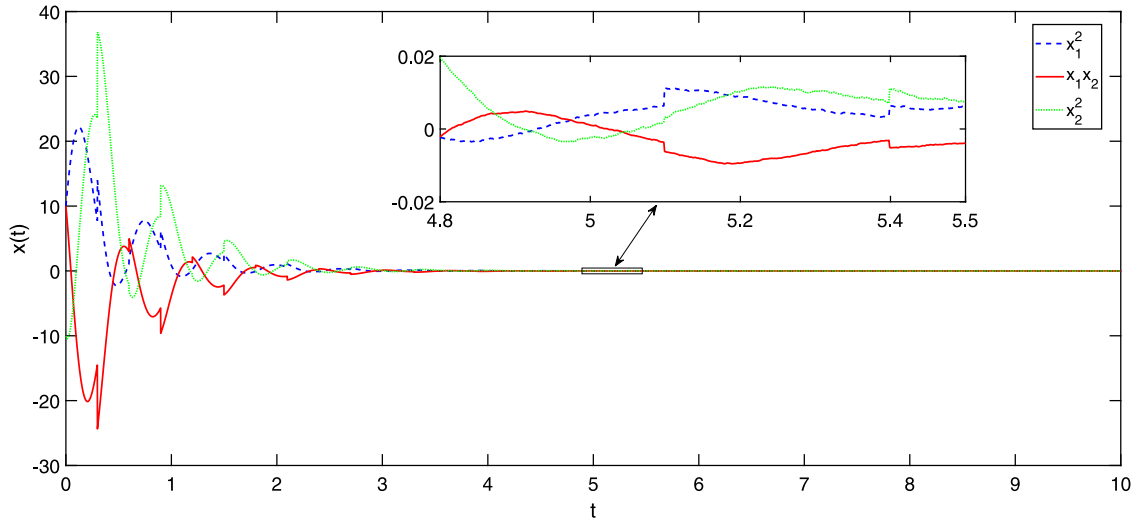
where

$$A_1 = \begin{bmatrix} 3 & 2 \\ -5 & 0 \end{bmatrix}, \quad J_1 = \begin{bmatrix} 1.35 & 0 \\ 0 & 1.25 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 3.1 & 2 \\ -4.8 & 0.1 \end{bmatrix}, \quad J_2 = \begin{bmatrix} 1.3 & 0 \\ 0 & 1.25 \end{bmatrix},$$

$$G_{w_1} = G_{w_2} = \begin{bmatrix} 0.03 & 0.02 \\ -0.024 & -0.001 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & -0.5 \\ 2 & -1 \end{bmatrix}.$$

$\eta_1(\epsilon(t)) = 1 - \cos^2(x_1(t))$, $\eta_2(\epsilon(t)) = \cos^2(x_1(t))$, which are shown in Fig. 1. Set the initial state to $x_1(t_0) = 10$, $x_2(t_0) = -10$ ($p = 1$). $x_1^2(t_0) = 10$, $x_1x_2(t_0) = 10$, $x_2^2(t_0) = -10$ ($p = 2$). $x_1^3(t_0) = 10$, $x_1^2x_2(t_0) = 10$, $x_1x_2^2(t_0) = -10$, $x_2^3(t_0) = -10$ ($p = 3$).

The state responses under $u(t) = 0$ ($p = 1$) are shown in Fig. 2.

Fig. 6. States responses ($p = 1$) with controller.Fig. 7. States responses ($p = 2$) with controller.

Remark 8. Stochastic systems are asymptotically stable in mean square stable at $p = 1$. Furthermore, higher-order moment stability guarantees lower-order moment stability; lower-moment is unstable higher-order moment must be unstable, but the converse is not true.

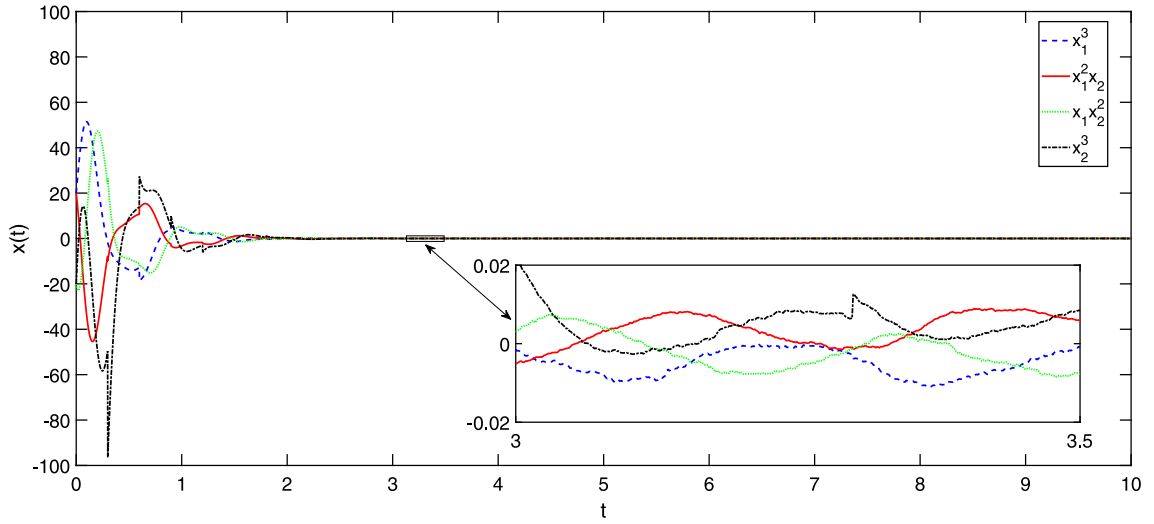
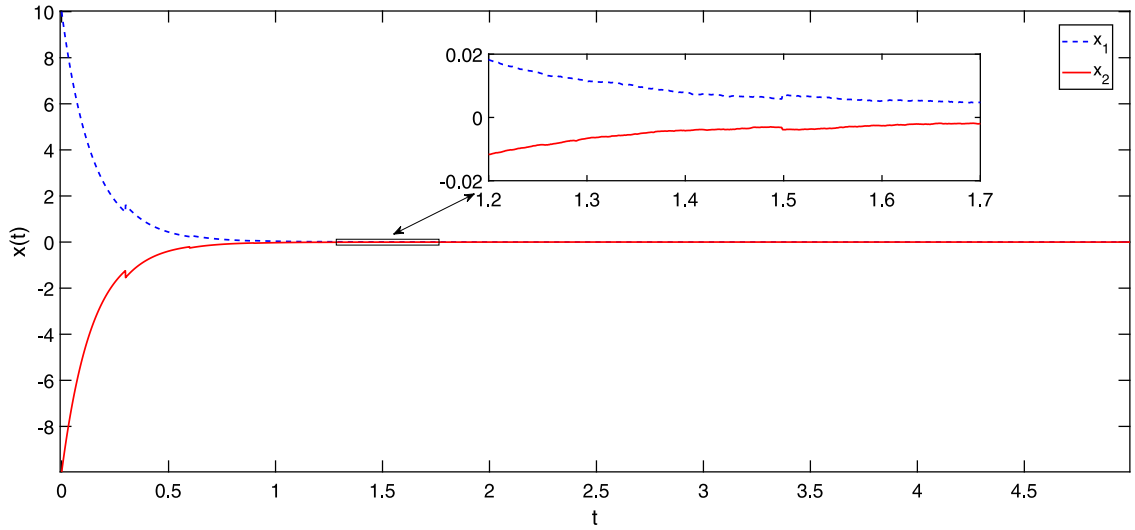
To compute the system's gain matrix, we can use [Corollary 2](#) and find it as follows

$$K_1 = \begin{bmatrix} -3 & -5 \\ 4 & 0 \end{bmatrix}, K_2 = \begin{bmatrix} -1 & -4 \\ 5 & -1 \end{bmatrix}.$$

The state responses of the system with controller are shown in [Figs. 3–5](#). When $p = 1$, $t > 3.4(s)$, system (26) converges to equilibrium; $p = 2$, $t > 1.9(s)$, system (26) converges to equilibrium; $p = 3$, $t > 1.2(s)$, system (26) converges to equilibrium.

The higher-order moment interval stability/stabilization technique proposed in this article is founded on the idea of pole structure. By adjusting the values of a and b , states responses of the system can be fine-tuned to converge at a specific rate towards the equilibrium point. In accordance with [Theorem 2](#), it is possible to regulate the convergence rate of the system by choosing the system's generalized eigenvalues at suitable interval locations. Executing the *Matlab* toolbox with $(-a, -b) = (-3, -1)$, the feasible solution is derived from the inequality in [Theorem 2](#) and their corresponding gain matrices are as follows

$$K_1 = \begin{bmatrix} -1 & -3 \\ 1 & 1 \end{bmatrix}, K_2 = \begin{bmatrix} -2 & -4 \\ 3 & -1 \end{bmatrix}.$$

Fig. 8. States responses ($p = 3$) with controller.Fig. 9. States responses ($p = 1$) with controller.

which can make convergence slower than above. The state responses with the controller are shown in Figs. 6–8. When $p = 1$, $t > 7.7(s)$, system (26) converges to equilibrium; $p = 2$, $t > 4.8(s)$, system (26) converges to equilibrium; $p = 3$, $t > 3(s)$, system (26) converges to equilibrium.

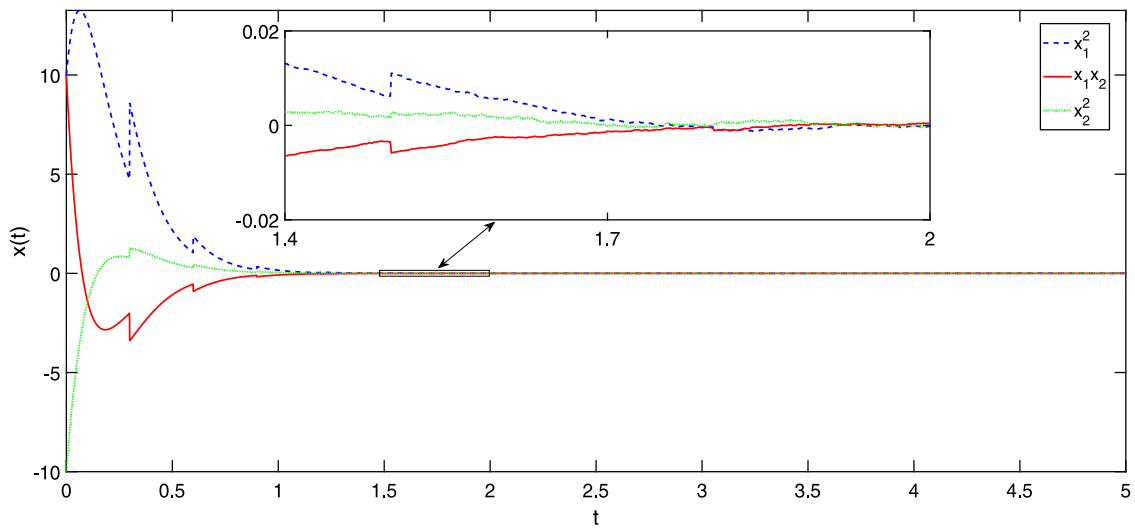
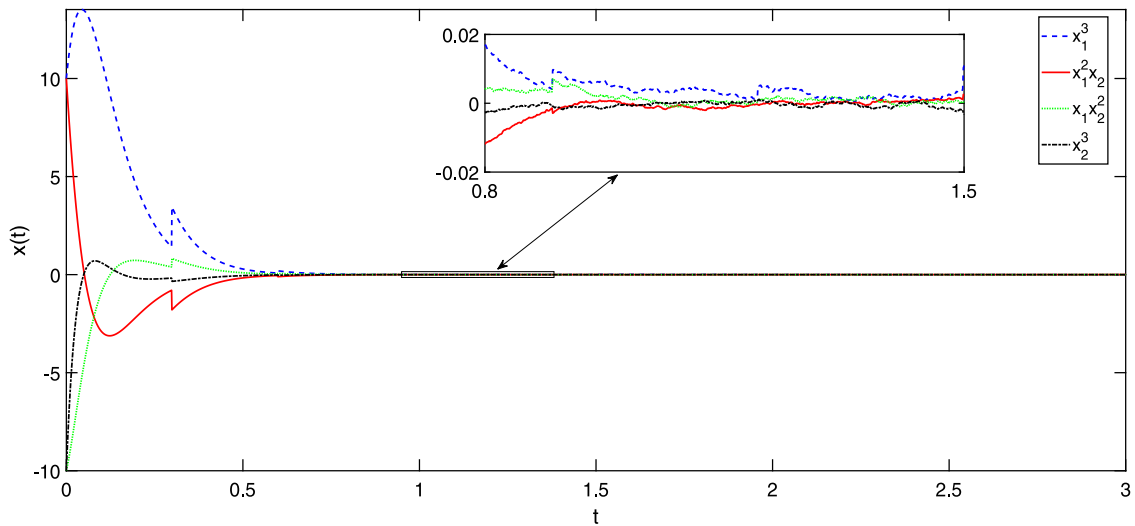
At the same time, adjusting the interval $(-a, -b) = (-8, -4)$ can be capable of increasing the speed at which the system converges to a balanced state. By referring to Theorem 2, the relevant gain matrices can be derived as

$$K_1 = \begin{bmatrix} 2 & -5 \\ 3 & 1 \end{bmatrix}, K_2 = \begin{bmatrix} 3 & -4 \\ 5 & 2 \end{bmatrix}.$$

which allows for faster convergence than above. The state responses with controller are shown in Figs. 9–11. When $p = 1$, $t > 1.2(s)$, system (26) converges to equilibrium; $p = 2$, $t > 1.4(s)$, system (26) converges to equilibrium; $p = 3$, $t > 0.8(s)$, system (26) converges to equilibrium.

6. Conclusion

In this paper, sufficient conditions are given for $2p$ th moment asymptotic interval stability of SITSFSs. Then, the proposed approach in combination with fuzzy control theory, the designed interval state controller can ensure the stability of the system

Fig. 10. States responses ($p = 2$) with controller.Fig. 11. States responses ($p = 3$) with controller.

while effectively regulating the convergence rate of the system. Meanwhile, high-order moment observability of SITFSFs also has been addressed. It believes that some problems in the high-order moment sense of SITFSFs, such as $2p$ th moment region stability, $2p$ th moment pole placement, etc., deserve investigation in future work.

CRediT authorship contribution statement

Tingting Zhang: Writing – original draft, Investigation, Formal analysis. **Huasheng Zhang:** Methodology, Conceptualization. **Xiangpeng Xie:** Supervision, Project administration. **Mengmeng Jiang:** Writing – review & editing, Validation.

Declaration of competing interest

The authors affirm the absence of any conflicts of interest.

Data availability

No data was used for the research described in the article.

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